1.5 A catalogue of essential functions

A polynomial P of degree n is given by $P(x) = a_n x^n + a_{n-1} x^{n-1} + a_{n-2} x^{n-2} + \dots + a_0$

where a_i, $i = 0, 1, \dots, n$ are arbitrary constants and $a_n \neq 0$, $n \in \mathbb{N}$

A rational function is given by

$R(x) = \dfrac{P(x)}{Q(x)}$ with P, Q polynomials.

linear function : $f(x) = P(x)$ with $n = 1$

power function : $f(x) = x^a$, $a \in \mathbb{R}$

Exponential functions : $f(x) = a^x$, $a > 0$

Logarithmic functions : $f(x) = \log_a x$, $a > 0$

Trigonometric functions : $\sin x$, $\cos x$, $\dots$

The graphs must be Known !!!